Linear Maps.

Linear Maps = Linear transformations = homomorphisms of vector spaces

Def. (15.a.1) Let V, W be vector spaces over K.

A map $T: V \longrightarrow W$ is called a linear map if:

1) $\forall\, v_1, v_2 \in V, \quad T(v_1 + v_2) = T(v_1) + T(v_2)$.

2) $\forall\, a \in K,\ v \in V, \quad T(a \cdot v) = a \cdot T(v)$.

Sometimes we write Tv insted of $T(v)$.

This means: $T: V \to W$ is a map that RESPECTS the structures of vector spaces on $V\ (+,\cdot)$ and $W\ (+,\cdot)$.

We denote the set of all linear maps $T: V \longrightarrow W$ by $\mathrm{Hom}_K(V,W)$ or $\mathrm{Hom}(V,W)$. Sometimes $\mathrm{Lin}(V,W)$, etc.
(or $\hom_K(V,W)$ or $\hom(V,W)$)

most std. notation.

Exc. $T: V \longrightarrow W$ is linear iff $T(a_1 v_1 + a_2 v_2) = a_1 T(v_1) + a_2 T(v_2)$

$\forall\, a_1, a_2 \in K,\ v_1, v_2 \in V$.

Examples. 1) $id_V : V \longrightarrow V, \quad id_V(v) := v \quad \forall\, v \in V$.

2) $0 : V \longrightarrow V, \quad 0(v) := 0 \quad \forall\, v \in V$.

Two obvious but important examples

3) $D : K[x] \longrightarrow K[x]$. The derivative.

$$D(a_0 + a_1 x + \ldots + a_k x^k + \ldots + a_n x^n) := a_1 + 2a_2 x + \ldots + k a_k x^{k-1} + \ldots + n a_n x^{n-1}.$$

$$D(a p(x) + b q(x)) = a \cdot p'(x) + b q'(x) = a D(p(x)) + b D(q(x))$$

$\forall\, a, b \in K,\ p(x), q(x) \in K[x]$.

3) $C([a,b]) :=$ continuous functions $f:[a,b]\to\mathbb{R}$.

$T: C([a,b]) \longrightarrow \mathbb{R}$. $\quad T(f) := \int_a^b f(x)\,dx$.

4) $S: K^\infty \longrightarrow K^\infty$, $\quad S((a_1,a_2,\dots,a_k,\dots)) := (a_2,a_3,\dots,a_k,\dots)$

The shift map.

Very important example. Let $A \in M_{m\times n}(K)$ be a matrix.

$T_A : K^n_{col} \longrightarrow K^m_{col}$,

$$T_A\left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\right) := \underset{m\times n}{A} \cdot \underset{n\times 1}{\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}} \in K^m$$

We have already seen that $A\cdot(a\cdot x) = a\cdot A\cdot x$ and $A(x+y) = Ax + Ay$ $\Rightarrow$ T_A is a linear map.

Prop. (15.a.2) Let V, W be vector spaces over a field K, and let $T: V \longrightarrow W$ be a linear map. Then:

1) $\forall\, n\in\mathbb{N}$, $a_1,\dots,a_n \in K$, $v_1,\dots,v_n \in V$ we have
$$T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T(v_i).$$

2) $T(0_V) = 0_W$.

Proof. 1) Follows from the def. of lin. map **(15.a.1)** + induction on n.

2) $0_W = T(0_V) - T(0_V) = T(0_V - 0_V) = T(0_V)$. ∎

Thm. (15.a.3) Let V be a finite dim. vector space and $v_1, \dots, v_n \in V$ a basis for V. Let $w_1, \dots, w_n \in W$ be arbitrary vectors. Then $\exists!$ a linear map $T: V \longrightarrow W$ s.t.

$$T(v_i) = w_i \quad \forall\, 1 \leq i \leq n.$$

Proof. Existence of T: Let $v \in V$. We want to define $T(v)$. Since $v_1, \dots, v_n$ is a basis, $\exists\ a_1, \dots, a_n \in K$ s.t. $v = a_1 v_1 + \dots + a_n v_n$. We define $T(v) := a_1 w_1 + \dots + a_n w_n$.

Note that T is well defined, b.c. $\forall\ v$ the coeffs. $a_1, \dots, a_n$ of the linear combin. $a_1 v_1 + \dots + a_n v_n$ that gives v are unique.

We claim that T is linear. Indeed, let $v, v' \in V$.

$\Rightarrow \exists\ a_1, \dots, a_n \in K$ s.t. $v = a_1 v_1 + \dots + a_n v_n$

and $\exists\ a_1', \dots, a_n' \in K$ s.t. $v' = a_1' v_1 + \dots + a_n' v_n$

$\Rightarrow v + v' = (a_1 + a_1') v_1 + \dots + (a_n + a_n') \cdot v_n$

$\Rightarrow$ by our def. of T, $T(v + v') = (a_1 + a_1') w_1 + \dots + (a_n + a_n') v_n =$

$= (a_1 w_1 + \dots + a_n w_n) + (a_1' w_1 + \dots + a_n' w_n) = T(v) + T(v')$.

Also, if $\alpha \in K$, then $\alpha \cdot v = \alpha a_1 v_1 + \dots + \alpha a_n v_n$

$\Rightarrow T(\alpha v) = \alpha a_1 w_1 + \dots + \alpha a_n w_n = \alpha (a_1 w_1 + \dots + a_n w_n) = \alpha T(v)$.

This proves that $T: V \longrightarrow W$ is a linear map.

We claim that $T(v_i) = w_i \ \forall\ 1 \leq i \leq n$. Indeed,

$v_i = 0 \cdot v_1 + \dots + 1 \cdot v_i + \dots + 0 \cdot v_n \Rightarrow T(v_i) = 0 \cdot w_1 + \dots + 1 \cdot w_i + \dots + 0 \cdot w_n = w_i$.

This concludes the proof of the existence of T.

<u>Uniqueness:</u> Suppose $T, S : V \to W$ are two linear maps with $T(v_i) = S(v_i) = w_i \ \forall\ 1 \le i \le n$.

Let $v \in V$. We have to prove that $T(v) = S(v)$.

Write $v = a_1 v_1 + \ldots + a_n v_n$.

$$T(v) = T(a_1 v_1 + \ldots + a_n v_n) = a_1 T(v_1) + \ldots + a_n T(v_n) =$$

$$= a_1 S(v_1) + \ldots + a_n S(v_n) = S(a_1 v_1 + \ldots + a_n v_n) = S(v). \qquad \blacksquare$$

Important example. (15.a.4) Consider K^n with the std. basis $e_1, \ldots, e_n$. Let $w_1, \ldots, w_n \in K^m$ be arbitrary vectors.

By the prev. Thm. (15.a.3), $\exists!$ a lin. map $T : K^n \to K^m$ with $T(e_i) = w_i \ \forall\ i$. Q. Describe T more explicitly.

Ans. Consider the matrix $A \in M_{m \times n}$ whose cols. are $w_1, \ldots, w_n$:

$$A = \begin{pmatrix} | & | & & | \\ w_1 & w_2 & \cdots & w_n \\ | & | & & | \end{pmatrix}. \qquad \text{Consider } \begin{array}{rcl} T_A : K^n & \longrightarrow & K^m \\ \rotatebox{90}{$\in$} & & \rotatebox{90}{$\in$} \\ v & \longmapsto & A \cdot v \end{array}$$

We have
$$T_A(e_i) = \begin{pmatrix} | & & \overset{\text{(col. } i)}{|} & & | \\ w_1 & \cdots & w_i & \cdots & w_n \\ | & & | & & | \end{pmatrix} \cdot \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\text{(entry } i)} = \begin{pmatrix} | \\ w_i \\ | \end{pmatrix}.$$

Lemma. (15.a.5) For every lin. Transf. $T: K^n \longrightarrow K^m$ $\exists$ a matrix $A \in M_{m\times n}(K)$ s.t. $T = T_A$.

Proof. Define $w_i = T(e_i)$ $\forall$ $1 \leq i \leq n$. Since $e_1, \dots, e_n$ is a basis for K^n, we know from the prev. Thm. (15.a.3) that every linear transf. S that satisfies $S(e_i) = w_i$ $\forall$ $1 \leq i \leq n$, must coincide with T.

Take $A = \begin{pmatrix} | & & | \\ w_1 & \dots & w_n \\ | & & | \end{pmatrix} \in M_{m\times n}(K)$. We have seen that

$T_A(e_i) = w_i$ $\forall$ $1 \leq i \leq n$. $\Longrightarrow$ $T = T_A$. ∎